

Задача:

Реализовать keygen для генерации лицензионных ключей для приложения и patcher для полного отключения лицензирования

Выполнение работы и результаты:

1) Декомпилировал программу через Ghidra, чтобы посмотреть как работает создание ключей:

```
calc_md5(PSN, 0x10);  
for (local_34[5] = 0; (int)local_34[5] < 0x10; local_34[5] = local_34[5] + 1) {  
    sprintf(md5decode + (int)(local_34[5] * 2), "%02x",  
        (ulong)(byte)md5digest[(int)(0xf - local_34[5])]);  
}
```

2) Написал keygen.py и проверил на работоспособность:

```
namkitu@namkitu-IdeaPad:~/labs/lab2$ ./unpatched_app  
Welcome to Lab2 super secure program!  
Your HWID is 010F8600FFFB8B17.  
Enter the license key: key  
Provided key is wrong! App is closing!  
Press Enter to continue...  
namkitu@namkitu-IdeaPad:~/labs/lab2$ ./keygen.py 010F8600FFFB8B17  
a009048bab129a11c5c9514174c4551  
namkitu@namkitu-IdeaPad:~/labs/lab2$ ./unpatched_app  
Welcome to Lab2 super secure program!  
Your HWID is 010F8600FFFB8B17.  
Enter the license key: a009048bab129a11c5c9514174c4551  
Now you app is activated! Thanks for purchasing!  
Press Enter to continue...  
namkitu@namkitu-IdeaPad:~/labs/lab2$
```

3) Посмотрел декомпилированный код и нашел переменную, которую нужно поменять, чтобы пропустить проверку лицензии:

00101400	48 89 45 f8	MOV	qword ptr [RBP + local_10], RAX	18	local_34[8] = 0;
00101404	31 c0	XOR	EAX, EAX	19	local_34[4] = 0;
00101406	c7 45 ec	MOV	dword ptr [RBP + local_1c], 0x1	20	__get_cpuid(1, local_34, local_34 + 1, local_34 + 2, local_34 + 3);
0010140d	01 00 00 00			21	local_34[7] = local_34[0] << 0x18
0010140f	c7 45 d4	MOV	dword ptr [RBP + local_34], 0x3	22	local_34[0] >> 0x18 (local_34[0] & 0xff00) << 8 local_34[0] >> 8
00101414	c7 45 d8	MOV	dword ptr [RBP + local_30], 0x0	23	local_34[8] = local_34[3] << 0x18
0010141b	c7 45 dc	MOV	dword ptr [RBP + local_2c], 0x0	24	local_34[3] >> 0x18 (local_34[3] & 0xff00) << 8 local_34[3] >> 8
00101422	c7 45 e0	MOV	dword ptr [RBP + local_28], 0x0	25	snprintf(PSN, 0x11, "%08X%08X", (ulong)local_34[7], (ulong)local_34[8], in_R9, param_2);
00101429	c7 45 f0	MOV	dword ptr [RBP + local_18], 0x0	26	calc_md5(PSN, 0x10);
00101430	c7 45 f4	MOV	dword ptr [RBP + local_14], 0x0	27	for (local_34[5] = 0; (int)local_34[5] < 0x10; local_34[5] = local_34[5] + 1) {
00101437	c7 45 e4	MOV	dword ptr [RBP + local_24], 0x0	28	printf(md5decode + (int)(local_34[5] * 2), "%02x",
0010143e	48 8d 7d e0	LEA	param_1=>local_28, [RBP + -0x20]	29	(ulong)(byte)md5digest[(int)(0xf - local_34[5])));
00101442	48 8d 4d dc	LEA	RCX=>local_2c, [RBP + -0x24]	30	}
00101446	48 8d 55 d8	LEA	RDX=>local_30, [RBP + -0x28]	31	readlink("/proc/self/exe", binaryPath, 0x1000);
0010144a	48 8d 75 d4	LEA	param_2=>local_34, [RBP + -0x2c]	32	getxattr(binaryPath, "user.license", xattrValue, 0x1000);
0010144e	8b 45 ec	MOV	EAX, dword ptr [RBP + local_1c]	33	puts("Welcome to Lab2 super secure program!");
00101451	49 89 f8	MOV	R8, param_1	34	iVar1 = strcmp(md5decode, xattrValue, 0x21);
00101454	89 c7	MOV	param_1, EAX	35	if (iVar1 == 0) {
00101456	e8 c3 fe	CALL	__get_cpuid	36	local_34[4] = 1;
0010145b	8b 45 d4	MOV	EAX, dword ptr [RBP + local_34]	37	}
0010145e	c1 e8 18	SHR	EAX, 0x18	38	if (local_34[4] == 0) {
00101461	89 c2	MOV	EDX, EAX	39	printf("Your HWID is %08X%08X.\nEnter the license key: ", (ulong)local_34[7], (ulong)
00101463	8b 45 d4	MOV	EAX, dword ptr [RBP + local_34]	40	__isoc99_scanf(&DAT_0010208f, userInput);
00101466	c1 e0 08	SHL	EAX, 0x8	41	iVar1 = strcmp(md5decode, userInput, 0x21);
00101469	25 00 00	AND	EAX, 0xff0000	42	if (iVar1 == 0) {
0010146c	ff 00	JB	EBX, EAX	43	setxattr(binaryPath, "user.license", md5decode, 0x21, 0);
0010146e	8b 45 d4	MOV	EAX, dword ptr [RBP + local_34]	44	puts("Now you app is activated! Thanks for purchasing!");
00101471	c1 e8 18	SHR	EAX, 0x18	45	}
00101474	89 c2	MOV	EDX, EAX	46	else {
00101477	8b 45 d4	MOV	EAX, dword ptr [RBP + local_34]	47	puts("Provided key is wrong! App is closing!");
0010147a	c1 e0 08	SHL	EAX, 0x8	48	}
0010147d	25 00 00	AND	EAX, 0xff0000	49	}
0010147f	ff 00	JB	EBX, EAX	50	else if (local_34[4] == 1) {
00101481	8b 45 d4	MOV	EAX, dword ptr [RBP + local_34]	51	puts("Your app is licensed to this PC!");
00101484	c1 e8 18	SHR	EAX, 0x18	52	}
00101487	89 c2	MOV	EDX, EAX		

4) Написал patcher.py и проверил его работу:

```

namkitu@namkitu-IdeaPad:~/labs/lab2$ ./hack_app
Welcome to Lab2 super secure program!
Your HWID is 010F8600FFFB8B17.
Enter the license key: abc
Provided key is wrong! App is closing!
Press Enter to continue...
namkitu@namkitu-IdeaPad:~/labs/lab2$ ./patcher.py ./hack_app
File './hack_app' patched successfully
namkitu@namkitu-IdeaPad:~/labs/lab2$ ./hack_app
Welcome to Lab2 super secure program!
Your app is licensed to this PC!
Press Enter to continue...
namkitu@namkitu-IdeaPad:~/labs/lab2$

```

5) Теперь в декомпилированном виде программа выглядит так:

<pre> 00101377 04 48 80 MOV RAX,qword ptr [FS:[0x28]] 00101400 48 89 45 f8 MOV qword ptr [RBP + local_10] 00101404 31 c0 XOR EAX,EAX 00101406 c7 45 ec MOV dword ptr [RBP + local_1c] 0010140d c7 45 d4 MOV dword ptr [RBP + local_34] 00101414 c7 45 d8 MOV dword ptr [RBP + local_30] 0010141b c7 45 dc MOV dword ptr [RBP + local_2c] 00101422 c7 45 e0 MOV dword ptr [RBP + local_28] 00101429 c7 45 f0 MOV dword ptr [RBP + local_18] 00101430 c7 45 f4 MOV dword ptr [RBP + local_14] 00101437 c7 45 e4 MOV dword ptr [RBP + local_24] 0010143e 01 00 00 00 0010143e 48 8d 7d e0 LEA param_1=>local_28,[RBP + 00101442 48 8d 4d dc LEA RCX=>local_2c,[RBP + -0x2 00101446 48 8d 55 d8 LEA RDX=>local_30,[RBP + -0x2 0010144a 48 8d 75 d4 LEA param_2=>local_34,[RBP + 0010144e 8b 45 ec MOV EAX,dword ptr [RBP + loca 00101451 49 89 f8 MOV RB, param_1 00101454 89 c7 MOV param_1,EAX 00101456 e8 c3 fe CALL __get_cpuid 0010145b 8b 45 d4 MOV EAX,dword ptr [RBP + loca 0010145e c1 e8 18 SHR EAX,0x18 00101461 89 c2 MOV EDX,EAX 00101463 8b 45 d4 MOV EAX,dword ptr [RBP + loca 00101466 c1 e0 08 SHL EAX,0x8 00101469 25 00 00 AND EAX,0xff0000 0010146a ff 00 OR EDX,EAX </pre>	<pre> 16 local_34[3] = 0; 17 local_34[7] = 0; 18 local_34[8] = 0; 19 local_34[4] = 1; 20 __get_cpuid(1,local_34,local_34 + 1,local_34 + 2,local_34 + 3); 21 local_34[7] = local_34[0] << 0x18 22 local_34[0] >> 0x18 (local_34[0] & 0xff00) << 8 loc 23 local_34[8] = local_34[3] << 0x18 24 local_34[3] >> 0x18 (local_34[3] & 0xff00) << 8 loc 25 snprintf(PSN,0x11,"%08X%08X", (ulong)local_34[7], (ulong)local_34[8],ir 26 calc_md5(PSN,0x10); 27 for (local_34[5] = 0; (int)local_34[5] < 0x10; local_34[5] = local_34 28 sprintf(md5decode + (int)(local_34[5] * 2),"%02x", 29 (ulong)(byte)md5digest[(int)(0xf - local_34[5]))]); 30 } 31 readlink("/proc/self/exe",binaryPath,0x1000); 32 getxattr(binaryPath,"user.license",xattrValue,0x1000); 33 puts("Welcome to Lab2 super secure program!"); 34 iVar1 = strcmp(md5decode,xattrValue,0x21); 35 if (iVar1 == 0) { 36 local_34[4] = 1; 37 } 38 if (local_34[4] == 0) { 39 printf("Your HWID is %08X%08X.\nEnter the license key: ",(ulong)loc 40 __isoc99_scanf(&DAT_0010208f,userInput); 41 iVar1 = strcmp(md5decode,userInput,0x21); 42 if (iVar1 == 0) { 43 setxattr(binaryPath,"user.license",md5decode,0x21,0); 44 puts("Now you app is activated! Thanks for purchasing!"); 45 } 46 else { 47 puts("Provided key is wrong! App is closing!"); 48 } 49 } 50 else if (local_34[4] == 1) { 51 puts("Your app is licensed to this PC!"); 52 } </pre>
---	--

Переменная local_34[4] стала равна 1 вместо 0.

Программа keygen была реализована на python3:

Исходный код:

```
#!/usr/bin/env python3
```

```
import hashlib
```

```
import sys
```

```
def generate_key(hwid):
```

```
    try:
```

```
        if len(hwid) != 16 or not hwid.isascii():
```

```
            return "HWID must contains exactly 16 ASCII symbols"
```

```
        digest = hashlib.md5(hwid.encode()).digest()
```

```
        reversed_digest = digest[::-1]
```

```
        return reversed_digest.hex()
```

```
    except Exception as e:
```

```
        return f"Exception: {e}"
```

```
if __name__ == "__main__":
```

```
    if len(sys.argv) > 1:
```

```
        print(generate_key(sys.argv[1]))
```

```
    else:
```

```
        print("Example: python3 keygen.py <HWID>")
```

Программа patcher была реализована на python3:

Исходный код:

```
#!/usr/bin/env python3
```

```
import os
```

```
import sys
```

```
def patch_binary(file_path):
```

```
    if not os.path.exists(file_path):
```

```
        print(f"File not found: '{file_path}'")
```

```
        return
```

```
    try:
```

```
        with open(file_path, "r+b") as f:
```

```
            f.seek(0x143a)
```

```
            current_byte = f.read(1)
```

```
            if current_byte == b'\x00':
```

```
                f.seek(0x143a)
```

```
                f.write(b'\x01')
```

```
                print(f"File '{file_path}' patched successfully")
```

```
            elif current_byte == b'\x01':
```

```
                print(f"File '{file_path}' already patched")
```

```
            else:
```

```
                print(f"Expected 00 by offset {hex(0x143a)} but was  
{current_byte.hex()}")
```

```
            except Exception as e:
```

```
                print(f"Exception: {e}")
```

```
if __name__ == "__main__":
```

```
    if len(sys.argv) > 1:
```

```
        patch_binary(sys.argv[1])
```

```
    else:
```

```
        print("Example: python3 patcher.py <path>")
```